

Who uses PyTorch

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Company	Products/Services Using PyTorch	Description of Usage
Meta Platforms (Facebook)	- Facebook App - Instagram - Meta AI Research Projects	Developed PyTorch and uses it extensively for computer vision, natural language processing, and AI research across its platforms.
Microsoft	- Azure Machine Learning - Bing Search - Office 365 Intelligent Features	Integrates PyTorch into Azure services for AI development; employs PyTorch in search relevance, productivity tools, and various AI applications.
Tesla	- Autopilot System - Full Self-Driving (FSD) Capability	Uses PyTorch for training deep neural networks in computer vision and perception tasks critical for autonomous driving systems.
OpenAI	- GPT Models - DALL-E - ChatGPT	Utilizes PyTorch for training large-scale language models and generative models in natural language processing and computer vision.
Uber	- Uber Ride-Hailing Platform - Uber Eats Recommendations	Employs PyTorch for demand forecasting, route optimization, and developed Pyro, a probabilistic programming language built on PyTorch.

PyTorch

→ open src DL Library used to make powerful neural N/Ws

2002 → Scientific Computing framework called Torch

- Using torch, you can perform tensor based operations. You could use GPUs to perform tensor based operations.
- DL researchers wrote implementations of many neural N/Ws in Torch.

Torch Issues

Cons: ① Torch was a Lua-based framework. Torch framework was written in Lua programming lang.

Meta AI identified this issue, they completely created a new library that has all the powerful capabilities of Torch & we can code that library in Python.

② Computation graphs used in Torch were static. Once you build your model, a Comp Graph is created from it & that Comp graph is fixed & cannot be changed.

Torch + Python $\rightarrow$ PyTorch

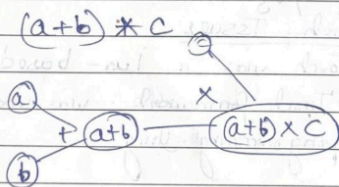
Flexibility $\uparrow$
Debugging Efficiency $\uparrow$

PyTorch Timeline

PyTorch 0.1 (2017):

Key Features:

- ① Python Compatibility: PyTorch was compatible with other Python Libraries. Eg NumPy, SciPy
- ② Dynamic Computation Graph



NNs are also mathematical ops.

PyTorch store & represent NN in form of Comp Graph. $(w \cdot x + b)$ $\rightarrow$ PyTorch store these ops in terms of graphs called Computation Graph.

Dynamic Comp graph can be changed at run time.

PyTorch 1.0 (2018)

Key Features:

- ① Bridged the gap b/w research & production environment
- ② Introduced TorchScript for model serialization & Optimization. TorchScript is good for running PyTorch models without ~~PyTorch~~ PyTorch.
- ③ Caffe2 integration. $\rightarrow$ another DL library
 $\rightarrow$ used with large models
 $\rightarrow$ Prod grade Deployment capabilities

PyTorch was already good for research but it also improved for prod.

PyTorch 1.x Series:
Key Features: ① Support for distributed training

→ realized you can't train your model by storing all data on just one machine. you train your model on multiple machines. Machines will have to be trained on multiple GPUs.

② ONNX (Open NN Exchange):

→ open src format.

→ Till now PyTorch can only be used in PyTorch based model. Tensorflow can only be used in Tensorflow based model. There was no interoperability b/w libraries.

• If you save a model in ONNX format then your created/trained model can also be used by other libraries.

③ Introduced quantization for model efficiency & compression.

model → wts → nos → float (32-bit)
1 Million wts → 1M x 32 bit
↓
32 Mbit
~ 900 mb

Quantization is a technique in which rather than storing your wts in float, you use a lower precision datatype like ~~float~~ int.

④ Expanded ecosystem with (cv) torchvision, torchtex (NLP) & torchaudio (audio)

Impact:

• Inspired community libraries like PyTorch Lightning

Similar to
Keras

→ high level API

for Tensorflow

• Hugging face Transformers

→ Library for Transformers

PyTorch 2.0

- Features:
- Performance Optimizations
 - Optimized for modern MLs

Core features of PyTorch

1. Tensor Computations
(multi-Dimensional arrays)
2. GPU Acceleration
3. Dynamic Computation Graph
4. Automatic Differentiation

→ Most imp Algo to train NN is backpropagation

AutoGrad → Module in PyTorch to which automatically differentiates tensors

5. Distributed training

→ Utilizes multiple Machines & multiple GPU clusters

6. Interoperability with other libraries

PyTorch Vs TensorFlow

	PyTorch By FB	TensorFlow By Google	
Aspect	Python	Supported Multiple lang Python (Primary), C++, Java, JS & Swift ↳ iPhone	Verdict
Ease of Use	More intuitive & Pythonic syntax	Complex	PyTorch Wins
Deployment & Prod ⁿ	Offers TorchScript for model serialization	Strong Prod ⁿ & Deploy (TensorFlow Lite & TensorFlow Serving)	TensorFlow Wins

Performance	Competitive performance optimized through dynamic graphs, static graphs & introduce slight overhead; optimized with JavaScript & JIT	XLA Compiler (efficient for large-scale Models)	Tie
Community	Rapidly growing large & established by Google; strong academic community; rich ecosystem with libraries like TensorFlow & TorchVision	TensorBoard & MuggingFace & TorchVision	TensorFlow has broader industry ecosystem
High-level APIs	PyTorch Lightning & Fast.ai	Keras	TensorFlow Wins
Mobile & Deployment	PyTorch Mobile enables deployment on iOS & Android. Supports quantization	TensorFlow Lite supports Mobile & embedded Device. TensorFlow.js for Web Deployment	TensorFlow Wins

Preferred Domain	Research & Academia	In Industry & Prod ⁿ	Tie
Learning Curve	Easier to learn due to intuitive Design	Complex	PyTorch Wins
Customizability	Easier to implement Custom Layers	Custom ops are Complex	PyTorch Wins
Interoperability	Supports exporting models to ONNX Format	Interoperable through TensorFlow Hub & SavedModel; Supports ONNX with some limitation	PyTorch Wins
Deployment Tools	TorchServe for model serving	TensorFlow Services & TFX for ML pipelines	TensorFlow Wins

Supports distributed training

Parallelism & Distributed Training	torch.distributed enhanced by libraries like Horovod	tf.distribute Strategy ↓ optimized for large-scale	TensorFlow Wine
Pretrained Models	TorchVision, Hugging Face	TensorFlow Hub	TF

Conclusion:

- To study DL, LLM & GenAI then master PyTorch
- To build conventional DL Models in industry then master TensorFlow

Core PyTorch Modules

Module
① torch: Core module providing multi-dimensional arrays (tensors) & mathematical ops on them
② torch.autograd: Automatic differentiation engine that records operations on tensors to compute gradients for optimization
③ torch.nn: Provides neural N/w Library, including layers, activations, loss fns & Utilities to build DL Models
④ torch.optim: Contains optimization algs like SGD, Adam, RMSProp
⑤ torch.utils.data: Classes ↓ Dataset ↓ DataLoader To load any dataset & managing it

- Just in time
Compilation
- ⑥ torch.jit : For TorchScript
 - ⑦ torch.distributed : For distributed training across multiple GPUs & machines ; parallel computation
 - ⑧ torch.cuda : interfaces with NVIDIA Cuda to enable GPU acceleration for tensor computations & model training
 - ⑨ torch.backends : For performance tuning & allow control over backend libraries like cudnn, MKL
 - ⑩ torch.multiprocessing : Utilities for parallelism using processing
 - ⑪ torch.quantization : To reduce model size & improve inference speed
 - ⑫ torch.onnx : To generate ONNX compatible models

- PyTorch Domain Libraries
- CNNs
 - ① torchvision : Provides datasets, model architectures, & image transformations for CV
 - NLP
 - ② torchtext : For Vocabulary management & data preprocessing
 - ③ torchaudio : For audio processing tasks
 - ④ torcharrow : for tabular data
 - ⑤ torchserve : for deployment
 - ⑥ pytorch-lightning : High-level API for PyTorch
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| ① Uber | ④ OpenAI |
| ② Meta | ⑤ Tesla |
| ③ Microsoft | |

Popular PyTorch Ecosystem Libraries

- ① Hugging Face Transformers: Provides state-of-art pretrained models for NLP tasks like text classification, translation & QNA built on PyTorch
- ② Fastai: High level library built on top of PyTorch
- ③ PyTorch Geometric: For Graph Neural N/Ws
- ④ TorchMetrics: Metrics used in PyTorch
- ⑤ TorchElastic: Provides elasticity in resource management
- ⑥ Optuna: Hyperparameter optimization framework
- ⑦ Catalyst: Provides high-level features
- ⑧ AllenNLP: NLP research based library
- ⑨ Skorch: Scikit + PyTorch
↳ scikit learn compatible wrapper that allows the use of PyTorch models
- ⑩ PyTorch Forecasting: for time-series data
- ⑪ Tensorboard for PyTorch: for visualization of training metrics